MILITARY SPECIFICATION

BODY ASSEMBLY: 11744900
(PERISCOPE: M35E1)

This specification is approved for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a modular unit, for daylight operation of the Tank Gunner's M35E1 Periscope, herein referred to as "body assembly."

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Method

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DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester, Two-Station
F11732109	Gage Interchangeability, Maximum
F11732110	Gage Interchangeability, Minimum
F11744900	Body Assembly
F11746735	Fixture, Alighment

PACKAGING DATA SHEET

11744900	Packaging of Body Assembly: 11744900

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F11744900 and drawings pertaining thereto.

3.2 Interchangeability.- The body assembly shall be capable of insertion into and removal from Periscope: M35E1 (gages are specified in lieu of periscope in 4.6.7.10).

3.3 General Specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specification MIL-O-13830 and MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The body assembly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature, plus 60° to plus 90° Fahrenheit (F), after having been exposed and thermally stabilized at temperatures of minus 80° and plus 160°F.

3.4.2 Operating temperatures.- The body assembly shall meet the applicable requirements of 3.17 while exposed and thermally stabilized at ambient of -40° and +150°F.

3.4.3 Vibration "A".- Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. The vibration shall be sinusoidal as follows:

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a. At a frequency of 5 Hertz (Hz), the acceleration shall be .65 G (peak).

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .65 to 1.5 G's (peak).

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's (peak).

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's (peak).

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .65 G (peak).

f. At 5 Hz, the acceleration shall be .65 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. The body assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the body assembly shall meet the requirements of 3.2 and 3.5 through 3.17.4 inclusive.

3.4.4 Vibration "B".- Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (HZ), the acceleration shall be 1.0 G (peak).

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G(peak).

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G (peak).

d. At 5 Hz, the acceleration shall be 1.0 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minute ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes (± 3 minutes). The body assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the body assembly shall meet the requirements of 3.2 and 3.5 through 3.17.4 inclusive.

3.4.5 Shock.- The body assembly shall be subjected to 24 cumulative shocks consisting of three each at 100 G's, then 40 G's, applied in each direction along each of the two mutually perpendicular horizontal axes (all four sides). The body assembly shall also be subjected to six cumulative shocks consisting of three each of 90 G's, then 35 G's applied in one direction along the vertical axis (normal operating position of body

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assembly only). The shock pulse shall have a half sine wave. For 100 G's ± 10% and 90 G's ± 10% the shock pulse shall have a time duration of 3 ± .2 milli-seconds; for 40 G's ± 10% the shock pulse shall have a time duration of 18 ± 3 milliseconds. The body assembly shall show no evidence of physical failure subsequent to shock. Following the shock test, the body assembly shall meet the requirements of 3.2 and 3.5 through 3.17.4 inclusive.

3.5 Cleanliness.

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing thru either end.

b. Dirt.- There shall be no more than three (3) particles of dirt or foreign material (classed as dirt) on any objective or eyepiece assembly. The apparent size of any dirt particle shall not exceed 0.05 inch.

3.6 Sealing.- The body assembly shall show no evidence of leakage in excess of 0.2 pounds per square inch (psi) when subjected to an internal pressure of 5 ± 0.10 psi for a minimum of 2 hours. The interior atmosphere of the accepted body assembly shall contain nitrogen having a dew point at least as low as minus 25°F at atmospheric pressure. The body assembly shall then be sealed with the dry nitrogen specified to an internal pressure of between 0.25 and 0.50 psig.

3.7 Orientation.- The requirements of 3.8 to 3.17 inclusive shall be met with:

a. The clear filter positioned in the field of view.
b. An infinity target projected into lens 8289292.
c. The eyepiece focused on the image of the projected infinity target.

NOTE: The following references apply throughout this specification.

Reference Line "Y" See Drawing F11744900

Datum Plane "X" (Intersection of planes)

3.8 Parallax.- Parallax between the image of the projected target, and the image of a target at 1200 meters shall not exceed 0.1 mil by adjustment of eccentric 10516549.

3.9 Objective focusing range.- Rotation of eccentric 10516549 shall cause the objective assembly to travel a minimum of 0.020 inch in each direction from the position established in 3.8.

3.10 Collimation.

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3.10.1 Pre-Vibration/Pre-Shock setting.- This requirement shall be met prior to meeting the requirements of 3.4.3 through 3.4.5 inclusive. An incident line of sight which is originally perpendicular to Datum Plane "X" shall coincide, within 0.20 mil, with the image of an infinity target point which is projected into lens 8289292 parallel to Reference Line "Y" (see 3.11.1).

3.11 Collimation change.

3.11.1 Due to Vibration/Shock.- This requirement shall be met subsequent to meeting the requirements of 3.4.3. An incident line of sight which is originally perpendicular to Datum Plane "X" shall coincide, within 0.30 mil (including fixture error), with the image of an infinity target point which is projected into Lens 8289292 parallel to Reference line "Y".

3.11.2 Due to the changing of filters.- The requirements of 3.10.1 and 3.11.1 shall also be met with each of the three remaining filters positioned in the field of view.

3.12 Image tilt.- The image of a target line which is originally parallel to surface "X" and perpendicular to reference line "Y" shall emerge from the eyepiece parallel to surface "X" within 1.0 degree of arc.

3.13 Parallelism of target image and projected image.- The image of a target line which is originally parallel to surface X and perpendicular to reference line "Y" shall emerge from the eyepiece parallel within 2 degrees of arc to the image of an infinity target line which is originally perpendicular to reference line "Y" projected into lens 8289292.

3.14 Magnification Accuracy.- Using a line target with an angular subtense of 50.00 $\pm$ 0.05 mils at 1200 meters $\pm$ 100 meters and an infinity line target with an angular subtense of 73.25 $\pm$ 0.05 mils projected into lens 8289292, the angular subtenses of the corresponding images shall agree within 0.10 mils (including parallax) measured on the 1200 meter target when viewed thru the eyepiece. Parallax between the images should be minimized by adjustment of eccentric 10516549.

3.15 Dioptral adjustment.

3.15.1 Zero setting.- This requirement shall be met while observing the projected image through a dioptometer set for zero diopter. The diopter scale shall indicate zero within 0.25 diopter when the image of a target projected through the projection system is brought into focus.

3.15.2 Excursion range.- The diopter movement shall be adjustable through a minimum range of -3.0 to +3.0 diopters as read on the diopter scale.

3.15.3 Backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter as read on the diopter scale.

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3.16 Resolution.- This requirement shall be met with the clear filter positioned in the field of view. Resolution shall be 6.0 seconds of arc or less when viewing with an auxiliary telescope having magnification of at least 3.0 power. Resolution consists of detectable line structure and proper line count in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopters.

3.17 Operability.

3.17.1 Diopter movement.- The running torque required to rotate the diopter movement through the range specified in 3.15.2 shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2 the running torque shall not exceed 15 lb-ins.

3.17.2 Focusing lock.- This requirement shall be met with a 14 ounce-inch (oz-in.) torque applied to the locking screw on Clamp 10516556. There shall be no visible rotation of Eccentric 10516549 when a 48 oz-in torque is applied to the eccentric.

3.17.3 Objective focusing mechanism.- This requirement shall be met with the locking screw backed off a minimum of two full revolutions from the position established in 3.8. The running torque required to rotate Eccentric 10516549 through a minimum of one full revolution in each direction shall not exceed 32 oz-ins. When checked at standard temperature. When checked at the temperature conditions specified in 3.4.2 the running torque shall not exceed 50 oz-in.

3.17.4 Filter assembly.- This requirement shall be met at standard temperature and the temperature conditions specified in 3.4.2 Handle 10516536 shall be manually operable to position each of the four filters in the field of view.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirement as specified herein. Except as other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three body assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Body Assembly.- Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.3, the tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejected by item or lot.

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4.3.2.3 100% Inspection.- The following requirements and related tests shall be subjected to 100% inspection:

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Dioptral adjustment zero setting	3.15.1	4.6.7.8.1
2.	Resolution	3.16	4.6.7.8.4
3.	Interchangeability	3.2	4.6.7.10

TABLE I - CLASSIFICATION OF DEFECTS

CLASS
CRITICAL: NONE
MAJOR: AQL 0.65% Defective

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Collimation, Pre-vibration setting	3.10.1	4.6.3
102.	Vibration "B"	3.4.4	4.6.4
103.	Collimation change due to vibration	3.11.1	4.6.7.1
104.	Cleanliness	3.5	4.6.5
105.	Sealing	3.6	4.6.6
106.	Parallax	3.8	4.6.7.3
107.	Image tilt	3.12	4.6.7.4
108.	Parellelism of reticle and image	3.13	4.6.7.5
109.	Objective focusing range	3.9	4.6.7.6
110.	Dioptral adjustmentexcursion range	3.15.2	4.6.7.8.2
111.	Diopter backlash	3.15.3	4.6.7.8.3
112.	Collimation change due to the changing of filters	3.11.2	4.6.7.2
113.	Magnification accuracy	3.14	4.6.7.7
114.	Operability-diopter movement	3.17.1	4.6.7.9.1
115.	Operability-focusing lock	3.17.2	4.6.7.9.2
116.	Operability-objective focusing mechanism	3.17.3	4.6.7.9.3
117.	Operability-filter assembly	3.17.4	4.6.7.9.4

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special Sampling.

4.4.1 General.- One body assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and test in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable Drawings - Visual
302.	General Specification	3.3	MIL-F-13926, MIL-O-13830 - Visual

NOTE: The tests in Table II shall be conducted at a temperature between 60° and 90°F.

4.4.2 Environmental.- Three body assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in 4.3.2.3 and in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° to +160°F)	3.4.1	4.6.1
304.	Operability, diopter movement	3.17.1	4.6.7.9.1
305.	Operability, focusing lock	3.17.2	4.6.7.9.2
306.	Operability, objective focusing mechanism	3.17.3	4.6.7.9.3
307.	Operability, filter Assembly	3.17.4	4.6.7.9.4

NOTE: The tests for Characteristic 304 through 307 inclusive, shall be conducted at -40° and at +150°F.

4.4.3 Verification, follow-up.- Three body assemblies shall be randomly selected as samples from production lots at 25 percent, 50 percent and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table IV.

TABLE IV - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
308.	Collimation, Pre-Vibration setting	3.10.1	4.6.3
309.	Vibration "A"	3.4.3	4.6.4.1
310.	Collimation change due to Vibration	3.11.1	4.6.7.1
311.	Collimation Pre-Shock setting	3.10.1	4.6.3
312.	Shock	3.4.5	4.6.4.3
313.	Collimation change due to shock	3.11.1	4.6.7.1

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4.4.4 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special testing equipment to test sealing of the body assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance.- Special test equipment to determine performance of the body assembly shall conform to the requirements of Drawing F11746735 (Fixture, Alignment).

4.5.2.3.2.1 Interchangeability.- Maximum and minimum interchangeability gages shall conform to Drawings F11732109 and F11732110 respectively.

4.6 Test methods and procedures.

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4.6.1 Storage temperature.- This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note below) to -80°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see Note below) to +160°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the body assembly removed from the test chamber. The body assembly shall then be subjected to the visual tactile and audible examination of 4.6.4. The body assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures.- This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note in 4.6.1) to -40°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F, the body assembly shall be subjected to the tests in 4.7.9 to determine compliance with the applicable requirements of 3.17, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see Note in 4.6.1) to +150°F. The body assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the body assembly shall again be subjected to the test in 4.7.9 to determine compliance with the applicable requirements of 3.17, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see Note in 4.6.1) to room ambient temperature (+60° to +90°F) and the body assembly removed from the test chamber. Body assemblies which have passed the above tests shall be subjected to the test in 4.6.3.

4.6.3 Collimation, pre-vibration/pre-shock setting.- Prior to vibration/shock, the body assembly shall be positioned as in 4.6.7 to insure conformance to 3.7. While viewing through the body assembly eyepiece, obtain coincidence of the projected infinity target and the image of the 1200 meter target.

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4.6.4 Vibration/Shock.

4.6.4.1 Vibration "A".- This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the body assembly shall be subjected to the test procedure specified in 4.6.3 to determine compliance with 3.10.1. Subsequent to the test of 4.6.3, the body assembly shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and frequency range shall be as specified in 3.4.3. Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the body assembly along three mutually perpendicular axes while the body assembly is maintained in it's normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F11744900. The body assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Periscope: M35E1. The body assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the period specified in 3.4.3. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the body assembly shall be subjected to the test in 4.6.7.1. Subsequent to the test in 4.6.7.1, the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. Tactile examination shall be made for functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall then be subjected to the tests specified in 4.6.5 through 4.6.7.10 inclusive.

4.6.4.2 Vibration "B".- This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the body assembly shall be subjected to the test procedure specified in 4.6.3 to determine compliance with 3.10.1. Subsequent to the test of 4.6.3, the body assembly shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and frequency range shall be as specified in 3.4.4. Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the body assembly along three mutually perpendicular axes while the body assembly is maintained in it's

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normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F117474900). The body assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Periscope: M35E1. The body assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.4.4. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the body assembly shall be subjected to the test in 4.6.7.1. Subsequent to the test in 4.6.7.1, the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.4, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall then be subjected to the tests specified in 4.6.5 through 4.6.7.10 inclusive.

4.6.4.3 Shock.- This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this shock test will not be required when the complete periscope is subject to shock. Prior to performing this test, the body assembly shall be subjected to the test procedure specified in 4.6.3 to determine compliance with 3.10.1. The shock equipment and test procedure shall be in accordance with Procedure I, of Method 516.2 specified in MIL-STD-810, except conditions such as "G" values, time duration and wave form shock pulse shall be as specified in 3.4.5. The body assembly shall be clamped on the shock equipment by means of an appropriate adapter having the same interface features as Periscope: M32E1. The body assembly shall then be shocked in accordance with the test procedure specified above, except the application of shock along the vertical axis shall be in one direction only (normal operating position of the body assembly). Immediately following the shock test and prior to the examination for physical defects resulting from poor workmanship and materials, the body assembly shall be subjected to the test in 4.6.7.1. Subsequent to the test in 4.6.7.1, the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.5, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall then be subjected to the tests specified in 4.6.5 through 4.6.7.10 inclusive.

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4.6.5 Cleanliness.- The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. The body assembly shall conform to the requirements of 3.5.

4.6.6 Sealing.- This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the body assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the body assembly by means of the dew point tester. Interior atmosphere of the body assembly shall conform to the requirements of 3.6. Upon satisfactory completion of this portion of the test internal pressurizing of the body assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the body assembly with dry nitrogen gas, having a dew point at least as low as -25°F to 5 pounds per square inch (psi) in accordance with the procedure for internal testing on Drawing F8565556. Retain the applied pressure in the body assembly for the time duration specified in 3.6. Within the first 60 minutes of this time period, the diopter adjustment shall be rotated to the clockwise and counterclockwise stops, and with the locking screw on Housing backed off from its locked position, the Eccentric shall be rotated a minimum to two full revolutions in each direction. At the end of this time period, any evidence of leakage as indicated on the pressure gage of the testing equipment shall not exceed the tolerance specified in 3.6. Upon satisfactory completion of the sealing test, the body assembly shall then be flushed with the dry nitrogen gas specified above. The interior atmosphere of the body assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained within the body assembly, the body assembly shall then be sealed at the pressure specified in 3.6. Upon completion, readjust the eccentric to the position established in 4.6.7.3 and torque the locking screw as specified in 3.17.2.

4.6.7 Performance.- To determine compliance with 3.7, the following tests shall be performed with the complete body assembly positioned on the special test equipment specified in 4.5.2.3.2. Set-up of the test equipment and installation of the body assembly under test shall be performed in accordance with the set-up and calibration instructions contained on Drawing F11746735. Unless otherwise specified during these tests, collimator "A" shall project a target image adjusted for 1200 meters, collimator "B" shall project an infinity target; Eccentric 10516549 (for adjusting parallax) shall be adjusted for minimum parallax between the target image and projected infinity image.

4.6.7.1 Collimation change due to vibration/shock.- While observing the body assembly eyepiece, the target image shall be coincident with the image of the infinity target within the tolerance specified in 3.11.1.

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4.6.7.2 Collimation change due to changing of filters.- Perform this test in sequence to test in 4.6.7.1. Observe through the body assembly eyepiece with the clear filter positioned in the field of view. Introduction of the remaining three filters into the optical system shall meet the collimation requirements specified in 3.11.2.

4.6.7.3 Parallax.- Perform this test with the body assembly diopter adjustment knob adjusted for best focus of the infinity target. Parallax requirements shall be met when the eccentric is adjusted as specified in 3.8. While viewing through the body assembly eyepiece parallax will be recognized by any apparent displacement of the infinity target with respect to the intersecting point of the vertical and horizontal target line image when the observer's head is moved from side to side or up and down. Displacement between the infinity target and target image shall not exceed the tolerance specified in 3.8.

4.6.7.3.1 Parallax adjustment.- If adjustment is necessary, a standard type screw driver that properly fits the screw slot of the locking screw on Housing 8634490 and Eccentric 8205635 shall be utilized for performing this adjustment. Back off the locking screw a minimum of two revolutions from its locked position. Then while performing the test in 4.6.7.3, rotate the eccentric clockwise or counterclockwise until the parallax requirement specified therein has been met. When the adjustment is complete, apply the applicable torque value specified in 3.15.4 to the locking screw.

4.6.7.4 Image tilt.- This test shall be performed with the aid of the observation telescope positioned at the body assembly eyepiece. View through the observation telescope and establish one extremity of the telescope horizontal reticle line coincident to the horizontal target image line. The image of the horizontal target line shall be parallel to the telescope vertical line within the tolerance specified in 3.12 as measured on the observation telescope.

4.6.7.5 Parallelism of target image and projected images.- Perform this test while observing through the body assembly eyepiece with the unaided eye. Position one extremity of the vertical line of the infinity target coincident to the vertical target image line. The vertical line of the infinity target shall be parallel to the vertical target image line within the tolerance specified in 3.13, as measured on the target.

4.6.7.6 Objective focusing range.- This test shall be performed subsequent to the test in 4.6.7.3. A dial indicator, graduated in .001 of an inch, mounted on an appropriate stand, a supporting medium capable of receiving the dial indicator as well as the body assembly to be tested shall be utilized for this test. With the body assembly firmly secured on the supporting medium, position the dial indicator so that its contact point touches the face of the movable objective tube of the body assembly and set the dial to read zero with the indicating hand. Back off the locking screw of Housing a minimum of two full revolutions from its locked position. Starting from the zero reading of the dial indicator, rotate the

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eccentric until the objective assembly has traveled the minimum distance specified in 3.9, as measured on the dial indicator. Return the indicating hand of the dial indicator to zero by means of the eccentric and repeat this travel in the opposite direction. There shall be sufficient travel in each direction to meet the minimum requirements specified in 3.9, as measured on the dial indicator. Upon completion of this test, repeat the parallax adjustment procedure specified in 4.6.7.3.1.

4.6.7.7 Magnification accuracy.- Two line targets constructed and positioned as specified in 3.14 shall be utilized for this test. While viewing through the eyepiece, the infinity target shall be coincident with the 1200 meter target within 0.10 mils, in elevation, as measured on the 1200 meter target.

4.6.7.8 Dioptral adjustment.

4.6.7.8.1 Zero setting.- A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy of the zero diopter setting of the diopter scale. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the eyepiece of the body assembly. While viewing through the dioptometer, obtain best focus of the infinity target by adjusting the body assembly diopter knob. The diopter scale shall indicate zero within the tolerance specified in 3.15.1.

4.6.7.8.2 Excursion range.- Perform this test in conjunction with the test in 4.6.7.8.1. From the best focus of the infinity target established in 4.6.7.8.1, rotate the diopter adjustment knob to its clockwise stop, note the diopter setting on the scale, then rotate the diopter knob to its counterclockwise stop, note diopter setting on the scale. The diopter movement shall meet the requirements specified in 3.15.2, as measured on the diopter of the body assembly.

4.6.7.8.3 Diopter backlash.- Perform this test utilizing the dioptometer specified in 4.6.7.8.1. The dioptometer shall be zeroed and adjusted as specified therein. Starting from the stop position rotate the body assembly diopter adjustment knob clockwise to the "0" position without reversing direction of rotation. Position the dioptometer at the eyepiece and obtain best focus of the infinity target by adjustment of the dioptometer diopter scale. Record the dioptometer scale reading. Rotate the body assembly diopter adjustment knob clockwise to its stop position, then return it counterclockwise to the "0" position without overtravel. Again obtain best focus of the infinity target by adjustment of the dioptometer diopter scale. The difference between the first and second dioptometer diopter scale readings shall not exceed the value specified in 3.15.3.

4.6.7.8.4 Resolution.- The test equipment specified in 4.6.7 is not required for this test. Perform this test utilizing the dioptometer specified therein. A resolving power chart on a wall target representing the angular subtense for six seconds of arc and a supporting medium capable

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of receiving the dioptometer (as well as the body assembly to be tested) shall be a part of the test equipment. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and shall contain four line sets as shown in Figure I therein. Place the body assembly on the supporting medium and position the dioptometer at the eyepiece. While viewing through the dioptometer, obtain best focus of the numeral in the center of the resolving power chart by adjusting the body assembly diopter knob, then obtain best focus of each of the four meridans by adjustment of the dioptometer objective scale. Any change in the diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. All four meridans shall have the detectable line structure and proper line count to conform with 3.16.

4.6.7.9 Operability.

4.6.7.9.1 Diopter movement.- Standard torque measuring equipment capable of measuring the torque values specified in 3.17.1 shall be utilized for this test. An appropriate adapter, to accommodate the diopter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter movement through the range specified in 3.17.1. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.17.1 for the temperatures indicated.

4.6.7.9.2 Focusing lock.- This test shall be performed in sequence to the test in 4.6.7.9.1. A dial indicator, graduated in increments of .0001 of an inch, mounted on an appropriate stand, the supporting medium specified and a torque wrench capable of applying the specified torque values in 3.17.2, together with a standard type driver bit that properly fits the screw slot of the locking screw and eccentric, shall be utilized for performing this test. Secure the body assembly firmly on the supporting medium. With the specified torque applied to the locking screw, position the dial indicator so that its contact point touches the face of the movable objective tube of the daylight body and set the dial to read zero. Apply, in a clockwise direction, the specified torque to the eccentric and again observe the dial indicator reading. To conform to the requirements of 3.17.2, there shall be no change in the established dial indicato to zero reading when observed after each torque application.

4.6.7.9.3 Objective focusing mechanism.- Perform this test utilizing the test equipment specified, except the dial indicator is not required. The locking screw shall be backed off a minimum of two full revolutions from its locked position. Torque shall be measured while rotating the eccentric through the range specified in 3.17.3. The running torque required to rotate the eccentric shall conform to the applicable requirements of 3.17.3 for the temperatures indicated. Upon completion of this test, return the eccentric to the position in 3.8 and torque the locking screw as specified in 3.17.2.

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4.6.7.9.4 Filter assembly.- Manually operate the Handle, 10516536. The handle shall be capable of positioning each of the four filters into the field of view. The handle shall conform to the requirements of 3.17.4, for the temperatures indicated in 3.4.2.

4.6.7.10 Interchangeability.- This test shall be performed utilizing the maximum and minimum interchangeability gages in Drawing F11732109 and F11732110 respectively, to determine compliance with 3.18.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11744900.

6. NOTES

6.1 Intended use.- The Body Assembly: 11744900, is a daylight modular unit of the Tank Gunner's Periscope: M35E1 primarily used in the fire control system of the 105MM Gun Tank: M60 Series.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data sheet (see 5.1).
d. Applicable stock number.
e. Provisions for first article testing.

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